

Shengfeng Li

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Senior Software Engineer with **7 years of experience** in **web development** and **microservices architecture**, specializing in designing and implementing **scalable, high-performance systems** for enterprise solutions. Proficient in integrating **DevOps practices** to enhance system reliability and operational efficiency. Possesses **5 years of experience** in **Android application development**, with strong expertise in **Java, AWS, and modern web technologies**. Additionally, has **2 years of experience** in **machine learning**, with a research focus on **reinforcement learning** and its applications.

Work Experience

TiMaven LLC - Senior Software Engineer

BATON ROUGE, LA May 2019 - Present

- Designed and developed a **labor hour management platform** that integrates **cost tracking systems, equipment tracking, procurement workflows**, and **customizable reports**, managing **13,000 employees** and over **5,000 machines**. Improved operational accuracy, transparency, and inter-departmental coordination through automation and optimized workflows.
- Built an **HR management system** to streamline workflows and enhance recruitment processes. Integrated **Twilio** for real-time communication using sockets, enabling instant messaging between HR personnel and candidates. Incorporated **customizable reports** to provide actionable insights, improving decision-making and employee management efficiency.
- Led the development of a secure **Project & Contract (P&C) website**, integrating advanced document management and reporting tools to support enterprise-level project oversight and collaboration.
- Spearheaded the redevelopment of a **healthcare platform** managing operations for over **20 clinics**. Leveraged **autoscaling architecture**, efficient design principles, and appropriate tools to achieve a **50% cost reduction**, while ensuring compliance with strict **medical standards**.

Louisiana State University - Research Assistant

BATON ROUGE, LA Sep 2017 - May 2019

- Designed and developed **CodeWorld**, an intuitive website serving over **20 teachers** and **500 students**, enabling educators with limited technical backgrounds to create teaching materials using easy-to-use templates. The platform streamlined assignment creation, distribution, and grading workflows.
- Researched and prototyped an **automated grading system** using **machine learning (ML)** techniques, including **Natural Language Processing (NLP)** for text-based assignments and **classification algorithms** for structured responses, aiming to reduce grading time and improve consistency.
- Developed a Python-based application utilizing **reinforcement learning** to train AI agents for the **Gomoku (Five in a Row)** board game as part of a Master's graduate project. Designed and implemented the game board, rules, and learning framework, enabling two AI agents to compete and improve their strategies autonomously through self-play.
- Enabled seamless interaction between educators and students: teachers created assignments, students submitted responses, and the system facilitated grading and result distribution.

BCL - Android Senior Software Engineer

BEIJING, CHINA July 2012 - Apr 2014, Feb 2016 - July 2017

- Led a team of **6 engineers** to design and implement Android applications for **snack&beverage vending machines**, managing unique protocols, embedded systems, and diverse hardware. Built profiles for multiple screen sizes (**7-inch, 10-inch, 32-inch**), customized interfaces, and **payment integrations**, including QR codes, mobile payments, and cash. Ensured compatibility and reliable item dispensing upon transactions.
- Migrated the **Honda Navigation System** from an embedded system to Android. Designed APIs, developed a custom framework to simplify the transition, and created user-friendly interfaces for better functionality.
- Developed an Android application for a **movie ticket vending system** to promote and sell tickets. Features included real-time movie trailers, seat selection, ticket purchasing, and ticket printing upon transaction completion. Integrated multiple payment methods to enable seamless transactions.

Fujitsu - Andoird Senior Software Engineer

TOKYO, JAPAN Apr 2014 - Feb 2016

- Designed and implemented a **fingerprint and iris recognition login app** integrated into the Android ROM, optimizing configurations for diverse conditions, improving performance using **JNI**, and ensuring robust **security measures**.
- Developed multiple Android apps, including **Happy Email**, which blended modern technology with a legacy UI for traditional users, and **Neatly Clock Alarm**, a clean, user-friendly alarm app that became one of the most popular in the market.
- Automated log analysis by writing a **bash script** to process large log files, reducing manual labor from days to seconds and significantly improving efficiency.

Certifications

AWS Certified Cloud Practitioner

Dec 2024

AWS Certified Solution Architect - Professional

Jan 2025

Education

Louisiana State University, Baton Rouge, LA - M.S. Computer Science

Aug 2017 - May 2019 GPA 3.8/4.00

Beijing Forestry University, Beijing, CHINA - B.S. Materials Science & Technology

Sep 2008 - July 2012

Skills

Java, Python, JavaScript, Kotlin, Bash, Android, Flutter, Spring Boot, Hibernate, JSP, Thymeleaf, REST APIs, Docker, Kubernetes, Git, CI/CD, PostgreSQL, MongoDB, Redis, Firebase, AWS (Lambda, EC2, S3, RDS, CloudFormation, EKS), Azure, GCP, Microservices, Domain-Driven Design, Event-driven Architecture, Autoscaling, Twilio Integration, Machine Learning, NLP, Embedded Development, Shell Scripting